

34.6 Summary and Notes

34.6.1 Summary

Idea.

1. An exponential dispersion family has density $\exp[w\{y\theta - b(\theta)\}/\phi + c(y, \phi, w)]$ with natural parameter θ , cumulant function b , dispersion ϕ and known weight w (Definition 34.1.1). The natural parameter space is convex and b is strictly convex on it (Lemma 34.1.1).
2. All moments come from b : $\mu = b'(\theta)$, $\text{Var}(Y) = \phi b''(\theta)/w$, and the r th cumulant is $(\phi/w)^{r-1} b^{(r)}(\theta)$ (Proposition 34.1.2). Reparameterizing by the mean gives the variance function $V(\mu)$ (Definition 34.1.2), which determines the family: $V = 1, \mu, \mu^2, \mu^3, \mu(1 - \mu), \mu(1 + \mu/k)$.
3. A generalized linear model is a random component from such a family, a linear predictor $\mathbf{X}\boldsymbol{\beta}$, and a link $g(\mu_i) = \eta_i$ (Definition 34.2.1); the canonical link $g = (b')^{-1}$ makes $\theta_i = \eta_i$ (Definition 34.2.2). The link and the variance function are separate decisions, and neither transforms the data.
4. The score is $\phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})$ with working weights $\mathbf{W}_{ii} = w_i h'(\eta_i)^2 / V(\mu_i)$; the expected information is $\phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$, and under the canonical link the observed information equals it (Theorem 34.3.1). The likelihood equations do not involve ϕ .
5. Under the canonical link the fit reproduces the weighted column totals of $\mathbf{X}$ exactly (Corollary 34.3.2), and the log-likelihood is strictly concave, so the maximum is unique when it exists (Theorem 34.3.3). Existence can fail, as in separation.
6. Fisher scoring is weighted least squares on the working response $z_i = \eta_i + (y_i - \mu_i)g'(\mu_i)$ with weights $\mathbf{W}$, iterated (Theorem 34.4.1). It is Newton's method exactly when the link is canonical, which is why canonical-link fits converge quadratically and others linearly.
7. Under stated regularity conditions — notably $\max_i q_{ni} \rightarrow 0$, the leverage-like condition Eq. 34.5.1 — the estimate exists with probability tending to one, is consistent, and $\mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0)$ is asymptotically standard normal (Theorem 34.5.1 for the canonical link, Proposition 34.5.2 for the rest). Wald, score and likelihood ratio statistics are asymptotically equivalent and $\chi^2(q)$ (Proposition 34.5.3), but the Wald statistic is not invariant and can fall as the evidence grows (the Hauck–Donner effect).
8. The deviance is twice the gap to the saturated log-likelihood (Definition 34.5.1); every contribution is nonnegative (Proposition 34.5.4). Differences of deviances between nested models are χ^2 in the limit; the deviance itself is a goodness-of-fit statistic only for grouped data with large groups, and for ungrouped binary data it is a function of $\hat{\boldsymbol{\beta}}$ alone (Theorem 34.5.5).
9. The dispersion is estimated by $X^2/(n - p)$ (Eq. 34.5.6), which leaves $\hat{\boldsymbol{\beta}}$ unchanged. Pearson, deviance and Anscombe residuals (Definition 34.5.2) and the weighted hat matrix carry the diagnostics of [Chapter 20](#) across, and AIC applies as in [Chapter 29](#).

34.6.2 Notes and sources

The synthesis. Nelder and Wedderburn (1972) created the subject. Every piece existed before — probit analysis, logistic regression, log-linear models for contingency tables, the gamma model for survival times, weighted least squares — and the paper’s contribution was to see that the exponential family structure made all of them one algorithm. That is what let a single program, GLIM, fit them all, and what makes the code of [Section 34.4](#) family-agnostic. McCullagh and Nelder (1989) remains the definitive treatment; its chapter 2 is the source for the deviance and residual definitions used here, chapter 12 for the diagnostics.

Exponential dispersion families. The class of Definition 34.1.1, with ϕ a genuine second parameter rather than a nuisance folded into θ , is due to Jorgensen (1987), developed in Jorgensen (1997); that paper also classifies the power variance functions $V(\mu) = \mu^\xi$, the Tweedie families of Exercise (C2). The analytic facts behind Proposition 34.1.2 are in Widder (1941); the statistical apparatus built on them, including the convexity of the natural parameter space and the mean-value mapping, is Barndorff-Nielsen (1978) and Brown (1986); the existence criterion quoted after Example (Two ways the maximum fails to exist) is Barndorff-Nielsen’s.

Fitting. The scoring method is older than the subject: it appears in Fisher’s appendix to Bliss (1935) on probit analysis, as a device for fitting a dose–response curve by hand. Nelder and Wedderburn (1972) identified the general form and recognized it as weighted least squares. Green (1984) is the standard reference for iteratively reweighted least squares as a general estimation device and for the convergence issues of [Section 34.4](#). Existence and uniqueness link by link are settled by Wedderburn (1976), the source of the gamma/identity warning in [Section 34.3](#); that only the variance function enters the estimating equations, so one may drop the density, is Wedderburn (1974), which [Chapter 38](#) follows.

Asymptotics. The complete theory — consistency, asymptotic normality, and existence with probability tending to one — is Fahrmeir and Kaufmann (1985), under conditions considerably weaker than (G1)–(G4) and covering non-canonical links and stochastic regressors; Proposition 34.5.2 quotes what this book needs from it. The proof of Theorem 34.5.1 takes the shortcut the canonical link allows: the log-likelihood is concave and its Hessian nonrandom, so the argument reduces to the central limit theorem of Lemma 21.2.1 for the score, plus the convexity lemma of Pollard (1991), which makes concavity do the work usually done by uniform convergence arguments. The condition $\max_i q_{ni} \rightarrow 0$ is the generalized-linear-model form of the bounded-leverage condition used throughout the book, and $\sum_i q_{ni} = p$ is Proposition 6.8.1(b) in a new setting.

The three tests. Cox and Hinkley (1974, chapter 9) is the standard general account of their asymptotic equivalence. The failure of Example (The Hauck–Donner effect) was identified by Hauck and Donner (1977) and named after them; it is one case of a general phenomenon, since the Wald statistic uses a quadratic approximation to the log-likelihood centred at $\hat{\beta}$, worst exactly where the estimate is most extreme. None of this arises in the normal linear model, where the log-likelihood is exactly quadratic in β for fixed σ^2 , and Proposition 11.4.2 records the exact relationships that result.

The deviance and its limits. That the residual deviance of a binary logistic fit is a function of $\hat{\beta}$ alone, hence useless as a goodness-of-fit statistic, is in McCullagh and Nelder (1989, section 4.4). The regime in which the deviance *is* a goodness-of-fit statistic — groups fixed, group sizes growing — is sometimes called grouped or small-dispersion asymptotics, against the $n \rightarrow \infty$ asymptotics of Theorem 34.5.1. The two give different answers for the same statistic, the source of most of the confusion in practice, and [Chapter 35](#) returns to it with Proposition 35.4.1.

Anscombe residuals are from Anscombe (1953); the extension of the diagnostics of [Chapter 20](#), including the weighted hat matrix and one-step deletion approximations, is Pregibon (1981).

Textbook treatments. Agresti (2015, chapter 4) covers the same ground at a similar level and is the closest companion; its chapters 5–7 correspond to [Chapter 35](#), [Chapter 36](#) and [Chapter 37](#). Fahrmeir, Kneib, Lang and Marx (2021, section 5.4) handles the weight w and grouped data carefully, and is the source of the convention used here that y may be a group average. Dobson and Barnett (2018) is a gentler introduction; Rencher and Schaalje (2008, chapter 18) treats the subject briefly, as an extension of linear-model theory.

What comes next. [Chapter 38](#) keeps the score equations and drops the density, which handles data more variable than the family allows. [Chapter 39](#) takes up model building in practice and the Bayesian treatment, whose priors are the counterpart of the penalized likelihoods of [Chapter 27](#) and [Chapter 30](#). [Chapter 40](#) joins the subject to the mixed models of [Chapter 32](#), where the likelihood is no longer available in closed form.

34.6.3 References

- Agresti, Alan (2015). *Foundations of Linear and Generalized Linear Models*. Hoboken, NJ: Wiley.
- Anscombe, Francis J. (1953). Contribution to the Discussion of H. Hotelling's Paper. *Journal of the Royal Statistical Society, Series B* 15(2), 229–230.
- Barndorff-Nielsen, Ole (1978). *Information and Exponential Families in Statistical Theory*. Chichester: Wiley.
- Bliss, Chester I. (1935). The Calculation of the Dosage–Mortality Curve. *Annals of Applied Biology* 22(1), 134–167.
- Brown, Lawrence D. (1986). *Fundamentals of Statistical Exponential Families, with Applications in Statistical Decision Theory*. Hayward, CA: Institute of Mathematical Statistics.
- Cox, David R. and Hinkley, David V. (1974). *Theoretical Statistics*. London: Chapman and Hall.
- Dobson, Annette J. and Barnett, Adrian G. (2018). *An Introduction to Generalized Linear Models*. 4th edition. Boca Raton, FL: Chapman and Hall/CRC.
- Fahrmeir, Ludwig and Kaufmann, Heinz (1985). Consistency and Asymptotic Normality of the Maximum Likelihood Estimator in Generalized Linear Models. *The Annals of Statistics* 13(1), 342–368.
- Fahrmeir, Ludwig, Kneib, Thomas, Lang, Stefan and Marx, Brian D. (2021). *Regression: Models, Methods and Applications*. 2nd edition. Berlin: Springer.
- Green, Peter J. (1984). Iteratively Reweighted Least Squares for Maximum Likelihood Estimation, and Some Robust and Resistant Alternatives. *Journal of the Royal Statistical Society, Series B* 46(2), 149–192.
- Hauck, Walter W. and Donner, Allan (1977). Wald's Test as Applied to Hypotheses in Logit Analysis. *Journal of the American Statistical Association* 72(360), 851–853.
- Jorgensen, Bent (1987). Exponential Dispersion Models. *Journal of the Royal Statistical Society, Series B* 49(2), 127–162.
- Jorgensen, Bent (1997). *The Theory of Dispersion Models*. London: Chapman and Hall.
- McCullagh, Peter and Nelder, John A. (1989). *Generalized Linear Models*. 2nd edition. London: Chapman and Hall.
- Nelder, John A. and Wedderburn, Robert W. M. (1972). Generalized Linear Models. *Journal of the Royal Statistical Society, Series A* 135(3), 370–384.
- Pollard, David (1991). Asymptotics for Least Absolute Deviation Regression Estimators. *Econometric Theory* 7(2), 186–199.

- Pregibon, Daryl (1981). Logistic Regression Diagnostics. *The Annals of Statistics* 9(4), 705–724.
- Rencher, Alvin C. and Schaalje, G. Bruce (2008). *Linear Models in Statistics*. 2nd edition. Hoboken, NJ: Wiley.
- Rockafellar, R. Tyrrell (1970). *Convex Analysis*. Princeton, NJ: Princeton University Press.
- Wedderburn, Robert W. M. (1974). Quasi-Likelihood Functions, Generalized Linear Models, and the Gauss–Newton Method. *Biometrika* 61(3), 439–447.
- Wedderburn, Robert W. M. (1976). On the Existence and Uniqueness of the Maximum Likelihood Estimates for Certain Generalized Linear Models. *Biometrika* 63(1), 27–32.
- Widder, David V. (1941). *The Laplace Transform*. Princeton, NJ: Princeton University Press.